

Evaluating Cybersecurity Awareness and Phishing Vulnerability Among College Students

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Background

Cybersecurity threats have become increasingly prevalent as digital technologies expand into everyday life. One of the most common forms of cybercrime is phishing, where attackers attempt to deceive individuals into revealing sensitive information such as passwords or financial data. College students are particularly vulnerable due to frequent email usage, online platforms, and limited formal cybersecurity training.

Recent studies show that human error remains one of the largest contributors to security breaches, with phishing attacks accounting for a significant portion of successful cyber intrusions (Verizon, 2023). Research also suggests that while students may be aware of cybersecurity risks, they often lack the practical skills to identify and respond to threats (Sheng et al., 2010). Understanding how students perceive and respond to phishing attempts can help improve cybersecurity education and reduce risks.

This project aims to evaluate cybersecurity awareness among undergraduate students and measure their susceptibility to phishing attempts through survey data and simulated scenarios.

Methodology

This study will use a mixed-methods approach combining surveys and simulated phishing assessments:

- **Participant Recruitment:**
Recruit 30–50 undergraduate students from various majors at UCF.
- **Survey Data Collection:**
Participants will complete a questionnaire assessing:
 - Knowledge of cybersecurity concepts
 - Previous exposure to phishing attempts
 - Confidence in identifying threats

- **Phishing Simulation:**
Participants will be shown a series of mock emails (legitimate and phishing examples) and asked to identify whether each is safe or malicious.
- **Data Analysis:**
 - Quantitative analysis of correct vs. incorrect identifications
 - Correlation between self-reported knowledge and actual performance
 - Identification of common mistakes and patterns

All data will be anonymized and collected in compliance with ethical research standards.

Anticipated Outcomes

The project is expected to reveal gaps between perceived and actual cybersecurity awareness among students. It will likely show that while many students feel confident in identifying phishing attempts, their actual detection accuracy may be lower.

Significance / Use of the Project

This research will contribute to improving cybersecurity education strategies by identifying specific areas where students struggle. The findings can inform universities on how to design more effective training programs.

Intended Audience

- University faculty and administrators
- Cybersecurity educators
- Undergraduate students
- Researchers in cybersecurity and human-computer interaction

Project Benefits

- Provides insight into student cybersecurity behavior
 - Identifies weaknesses in phishing detection skills
 - Supports development of targeted educational interventions
 - Contributes to broader research on human factors in cybersecurity
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References (APA Format)

Sheng, S., Holbrook, M., Kumaraguru, P., Cranor, L. F., & Downs, J. (2010). Who falls for phish? A demographic analysis of phishing susceptibility. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 373–382.

Verizon. (2023). *2023 Data Breach Investigations Report*. Verizon Enterprise.
<https://www.verizon.com/business/resources/reports/dbir/>

Timeline (Spring 2027 Semester)

Dates	Tasks
Jan 1 – Jan 15	Develop survey and phishing simulation materials
Jan 16 – Jan 31	Submit for IRB/approval (if required)
Feb 1 – Feb 28	Recruit participants
Mar 1 – Mar 20	Conduct surveys and simulations
Mar 21 – Apr 5	Analyze data
Apr 6 – Apr 20	Interpret results and identify trends
Apr 21 – Apr 30	Prepare final report and presentation

Budget (On-Campus Project: ≤ \$750)

Item	Units and Cost	Total
Participant incentives	\$10 × 40 participants	\$400
Survey software (Qualtrics premium features, if needed)	Flat fee	\$100
Printing materials (consent forms, flyers)	Estimated	\$50
Data analysis software (SPSS student license)	\$100	\$100
Miscellaneous research supplies	Estimated	\$50
Total Requested \$700		
